



SDG Module 1

Sustainable Development Goals (Visvesvaraya Technological University)



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MODULE – 1

SUSTAINABLE DEVELOPMENT

1. Introduction to Sustainable Development

Sustainable Development is a holistic approach to development that aims to meet the needs of the present generation without compromising the ability of future generations to meet their own needs. The concept integrates **economic growth, social equity, and environmental protection** into a single framework.

The most widely accepted definition was given by the **Brundtland Commission (1987)**:

“Sustainable development is development that meets the needs of the present without compromising the ability of future generations to meet their own needs.”

Sustainable development emphasizes:

- Long-term planning
- Responsible use of natural resources
- Inclusive growth
- Environmental conservation
- Social justice and equity

Traditional development models focused mainly on economic growth, often ignoring environmental degradation and social inequality. Sustainable development emerged as a response to these limitations.

2. Evolution of the Concept of Sustainable Development

The idea of sustainable development evolved gradually through global discussions and international conferences:

2.1 Early Environmental Concerns

- 1960s–1970s: Growing awareness of pollution, deforestation, and resource depletion
- Publication of *Limits to Growth* (1972) by the Club of Rome

2.2 Stockholm Conference (1972)

- First global conference on the human environment
- Recognized the link between environment and development

2.3 Brundtland Commission (1987)

- Introduced the modern definition of sustainable development
- Highlighted intergenerational equity

2.4 Rio Earth Summit (1992)

- Adoption of Agenda 21
- Emphasis on sustainable development at global and national levels

2.5 Agenda 2030 and SDGs (2015)

- Adoption of **17 Sustainable Development Goals**
 - Focus on people, planet, prosperity, peace, and partnership
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3. Pillars of Sustainable Development

Sustainable development is based on **three interconnected pillars**, often called the **Triple Bottom Line**.

3.1 Economic Sustainability

- Continuous economic growth
- Efficient use of resources
- Employment generation
- Technological innovation
- Long-term financial stability

3.2 Social Sustainability

- Poverty eradication
- Education and healthcare access
- Gender equality
- Social justice
- Inclusive development

3.3 Environmental Sustainability

- Conservation of biodiversity
- Climate change mitigation
- Pollution control
- Sustainable resource management
- Renewable energy adoption

All three pillars must function together for true sustainability.

4. Economic Growth and Economic Development

4.1 Economic Growth

Economic growth refers to an increase in a country's output of goods and services over time, usually measured by GDP.

Characteristics:

- Quantitative concept
- Focus on income and output
- Short-term perspective

4.2 Economic Development

Economic development is a broader concept that includes improvement in living standards, health, education, and equality.

Characteristics:

- Qualitative concept
- Focus on human well-being
- Long-term perspective

4.3 Growth vs Development

Economic Growth	Economic Development
Increase in GDP	Improvement in quality of life
Quantitative	Qualitative
May increase inequality	Reduces inequality

Sustainable development emphasizes **development rather than mere growth**.

5. Continuing Poverty in a Growing Economy

Despite economic growth, poverty continues to exist due to structural and systemic issues.

5.1 Causes of Poverty

- Unequal distribution of income
- Lack of education and skills
- Unemployment and underemployment
- Social exclusion
- Weak governance

5.2 Multidimensional Poverty

Poverty is not only lack of income but also:

- Poor health
- Illiteracy
- Inadequate housing
- Lack of sanitation and clean water

5.3 Poverty and Sustainability

Poverty leads to:

- Overexploitation of natural resources
- Environmental degradation
- Increased vulnerability to climate change

Thus, poverty eradication is a core objective of sustainable development.

6. Environmental Threats to Sustainable Development

Rapid industrialization and urbanization have caused severe environmental problems.

6.1 Climate Change

- Rising global temperatures
- Melting glaciers
- Extreme weather events
- Sea-level rise

6.2 Biodiversity Loss

- Habitat destruction
- Species extinction
- Ecosystem imbalance

6.3 Pollution

- Air pollution
- Water pollution
- Soil contamination
- Plastic waste

6.4 Resource Depletion

- Overuse of fossil fuels
- Deforestation
- Water scarcity

Environmental degradation directly threatens human survival and economic stability.

7. Business as Usual (BAU) Model

The **Business as Usual** approach refers to conventional development practices that prioritize short-term economic gains.

7.1 Features of BAU

- Profit maximization
- High resource consumption
- Environmental neglect
- Social inequality

7.2 Impacts of BAU

- Climate change acceleration
- Ecological imbalance
- Social conflicts
- Long-term economic instability

BAU is unsustainable in the long run.

8. Sustainable Development Model

The sustainable development model focuses on balancing growth with responsibility.

8.1 Key Principles

- Intergenerational equity
- Intragenerational equity
- Resource efficiency
- Environmental protection
- Inclusive growth

8.2 BAU vs Sustainable Development

Business as Usual	Sustainable Development
Short-term focus	Long-term focus
Resource exploitation	Resource conservation
Environmental damage	Environmental protection
Social inequality	Social inclusion

9. Sustainable Development Goals (SDGs)

The **17 SDGs**, adopted by the United Nations in 2015, provide a global roadmap for sustainable development by 2030.

9.1 Key SDGs

- No Poverty
- Zero Hunger
- Good Health and Well-being
- Quality Education
- Gender Equality
- Clean Water and Sanitation
- Affordable and Clean Energy
- Climate Action

9.2 Importance of SDGs

- Universal framework
 - Integrates economic, social, and environmental goals
 - Encourages global partnerships
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10. Role of Urbanization and Smart Cities

Rapid urbanization presents both challenges and opportunities for sustainability.

10.1 Urban Challenges

- Overcrowding
- Pollution
- Infrastructure stress
- Energy demand

10.2 Smart Cities for Sustainable Development

According to Mishra et al. (2022), smart cities:

- Use ICT and digital technologies
- Improve resource efficiency
- Enhance governance and service delivery
- Promote sustainable urban living

Smart cities play a crucial role in achieving SDGs.

11. Importance of Sustainable Development

Sustainable development is essential for:

- Long-term economic stability
 - Environmental protection
 - Social harmony
 - Global peace and security
 - Survival of future generations
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12. Conclusion

Sustainable development is not an option but a necessity in the modern world. It provides a balanced approach to development by integrating economic growth, social equity, and environmental protection. Moving away from business-as-usual models toward sustainable practices is essential to ensure a healthy planet and prosperous society for present and future generations.

(Prepared with reference to: Ram Kumar Mishra et al. (2022), Springer; UN SDG Report 2020; United Nations (2018))